

Assignment 3: Code Correction & Quality Assurance Sheet

Course: Advanced C Programming Assignment: No. 3 Status: Completed & Verified

QUESTION NO.	LINE NO.	ERROR TYPE	ERROR DESCRIPTION	ERROR CORRECTION
Problem 1	14	Syntax Error	Missing semicolon (;) at the end of the variable declaration or statement inside the loop structure.	Append a semicolon at the end of the statement: <code>int total = 0;</code>
Problem 2	28	Memory Leak	Dynamic memory allocated via <code>malloc()</code> for the node pointer is not freed before exiting the function or losing scope.	Add a call to <code>free()</code> before returning: <code>free(node);</code>
Problem 3	42	Logic Error	Incorrect loop termination condition leading to an Off-By-One error (using <code><=</code> instead of <code><</code> on array bounds).	Change condition to strict less-than: <code>for (int i = 0; i < MAX_SIZE; i++)</code>
Problem 4	19	Type Mismatch	Using an incorrect format specifier <code>%d</code> in <code>printf()</code> to output a floating-point value (<code>double</code> or <code>float</code>).	Change the format specifier to <code>%f</code> or <code>%lf</code> : <code>printf("Result: %f ", dynamic_average);</code>
Problem 5	8	Runtime Risk	Attempting to read values using <code>scanf()</code> without passing the address of the target variable (missing <code>&</code> operator).	Prepend the address-of operator: <code>scanf("%d", &userInput);</code>
Problem 6	35	Null Pointer Dereference	Accessing a struct member via a pointer returned by <code>fopen()</code> without verifying if the pointer is <code>NULL</code> (file open failure).	Insert a conditional safety check: <code>if (filePtr == NULL) { return ERROR_CODE; }</code>
Problem 7	51	Undefined Behavior	Using a local uninitialized variable inside a mathematical computation or accumulation loop.	Initialize the variable explicitly to zero at declaration: <code>int productSum = 0;</code>
Problem 8	22	Buffer Overflow	Using the unsafe <code>gets()</code> function for string input, which does not check boundaries and risks smashing the stack.	Replace with the safe <code>fgets()</code> function: <code>fgets(buffer, sizeof(buffer), stdin);</code>

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Problem 9	12	Pre-processor Error	Missing header file include for standard library functions like <code>exit()</code> or <code>malloc()</code> .	Add the missing include directive at the top: <code>#include <stdlib.h></code>
Problem 10	64	Logic Error	Using the assignment operator (<code>=</code>) instead of the equality comparison operator (<code>==</code>) inside an <code>if</code> condition statement.	Change the assignment operator to the comparison operator: <code>if (status == SUCCESS)</code>

Compilation & Quality Note:

All identified bugs across the 10 code files have been successfully tabulated. Implementing the recommended code corrections will eliminate compiler warnings, prevent memory leaks, ensure robust runtime boundary checks, and guarantee deterministic code behavior.